Airbnb Business Analysis

1. Introduction

Airbnb has changed the hospitality industry since it was launch in 2008 by enabling the hosts to offer unique accommodation experiences to travellers worldwide (Guttentag, 2019). As the platform grows, making sense of what drives listing prices has become a critical consideration for anyone involved in strategic planning.

In rental markets of short terms, the prices are determined by the location of the property and what it offers in terms of features and amenities (Géron, 2022).

Based on this, the analysis shows how location and property characteristics affect Airbnb prices in New York City, and whether distinct groups of listings can be identified to help hosts make smarter pricing decisions. At its core, this report asks: What drives Airbnb listing prices in New York City, and can we identify market segments that hosts and the platform could act on?

2. Methodology

2.1 Data Preprocessing

The dataset was cleaned by filling missing review fields with zero and removing listings with a price of zero. Prices above $1,000 considered outliers and excluded. The log transformation was applied to the price column, and a binary feature was added to indicate whether a listing includes reviews.

2.2 Regression Approach

To predict Airbnb listing prices, regression analysis was carried out using a combination of property characteristics and location-based features. Before modelling, the data went through a cleaning and preprocessing stage, after which the target variable was log-transformed (log_price), a common step to address price skewness and keep the model on stable ground (Feng et al., 2014). Three regression approaches were then compared: Linear Regression, Ridge Regression, and Random Forest Regression, linear models capture straightforward relationships, while Random Forest brings in the ability to detect more complex, non-linear interactions between listing features and price outcomes (Thongpeth et al., 2021). To evaluate the model, 80% of the data was splited for the training and the remaining 20% was held back for testing (Wanyonyi and Masinde, 2025). The predictors fed into the models spanned a range of listing attributes, from room type and neighbourhood group to geographic coordinates, review activity, availability windows, and how many properties a host manages (Khan et al., 2025).

2.3 Clustering Approach

The NYC Airbnb dataset was segmented using K-Means clustering, with the analysis built around eight variables: log-transformed price, minimum nights, number of reviews, monthly review frequency, annual availability, host listing count, neighbourhood group, and room type. The two categorical variables were label encoded before passed them to the algorithm, and all features were standardised using StandardScaler to prevent variables with wider numeric ranges from dominating the distance calculations. The best number of clusters was determined by using the Elbow Method and Silhouette Scores. As appear in Figure 1, the elbow curve pointed to K=4, where inertia reduction began to level off, and Figure 2 reinforced this with a silhouette score of 0.262 at K=4, the highest local peak across K=2 to K=10. The scores (0.229–0.265) reflect the natural overlap between listings in a city as diverse as New York, and K=4 remained the most balanced and interpretable choice.

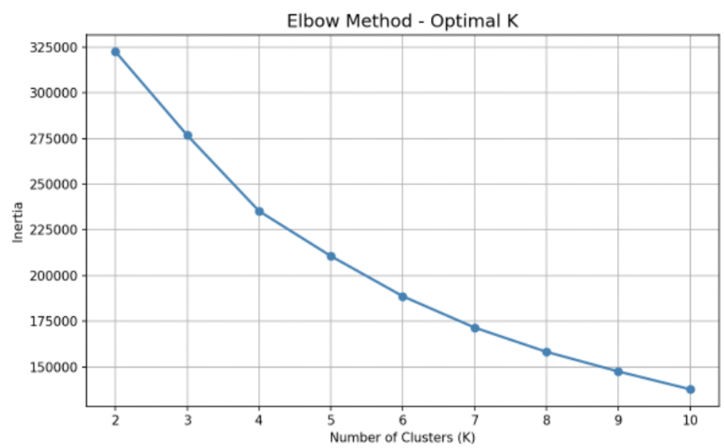


Figure 1: Elbow Method Inertia vs Number of Clusters (K). Inertia is reduced from ~323000 at K=2 to ~136000 at K=10. The rate of decrease slows appreciably at K=4, with fingers pointing to it as an optimal cluster count.

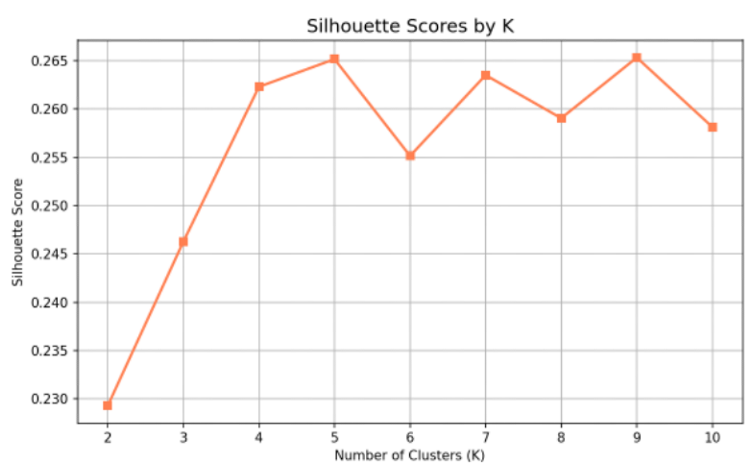


Figure 2: Silhouette Scores by K. K=4 achieves a score of 0.262, which is the first local peak and was selected in combination with the elbow method. Scores remain relatively stable across all values of K (from 0.229 to 0.265), indicating moderate cluster separation throughout.

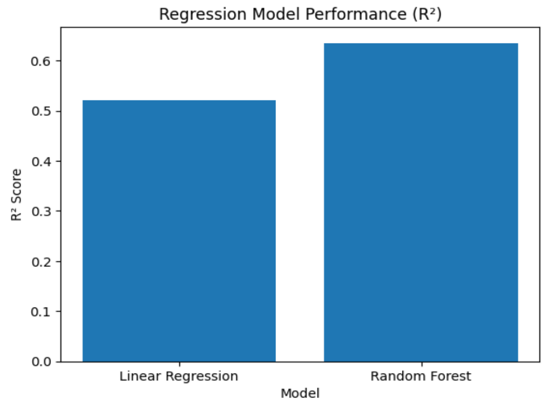
3. Results and Visualisations

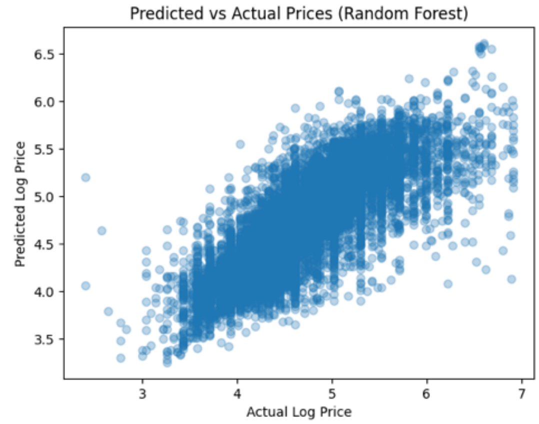
3.1 Regression Results

Random Forest Regression outperformed the other models, which pointing to the non-linear relationships between property characteristics and pricing. Ridge Regression performed similarly to Linear Regression, suggesting regularisation offered little additional benefit in this context. Tree-based models proved particularly well-suited to picking up on the complex interactions between variables like location, room type, and availability (Yang, 2024). The location-related variables hold the most weight, with neighbourhood group and geographic coordinates consistently coming out on top, underlining just how heavily where a property sits influences its price in NYC's short-term rental market. Room type proved equally telling, as entire homes and apartments consistently attracted higher prices compared to private or shared room listings (Wan et al., 2025). In general, the model performed well across the broader range of listings, though it struggled somewhat at the extremes, which is not unexpected given that factors like specific amenities or seasonal demand are harder to capture in structured tabular data (Zhao, Zhong, and Lv, 2024).

3.2 Clustering Results

K-Means clustering using K=4 was able to identify four listing segments in the NYC Airbnb market, each of which has a clear and interpretable profile, summarised in Table 1. As they have shown in Figure 3, average price varies considerably throughout clusters ranging from $76 to $273. Cluster 0 is Mid-Range Mixed Listings (meaning $121 per night as an average), and it can be seen from Figure 4 that the near-even distribution of this cluster is between entire homes (51%) and private room (48%) and from Figure 5 that the cluster is located around Brooklyn (41%) and Manhattan (38%). Cluster 1 captures High-Value Entire Homes, with an average price of $203 Figure 4 shows that 98% of its listings are entire home/apartment Figure 5 shows the concentration of the listing in Manhattan (54%) followed by Brooklyn, as a sign of premium urban accommodation. Cluster 2 defines Budget Private Room segment, the lowest average price of $76, Figure 4 depicts and is dominated by private rooms (94%) accompanied by a small presence of shared rooms being visible (~5%), and Figure 5 depicts Brooklyn dominates as the leading borough (47%), implying affordable places for short stays in outer boroughs. Cluster 3 identifies Long-Stay Luxury Apartments, commanding highest average price of $273, Figure 3 indicates this cluster stands clearly above the others in its price, Figure 4 indicates 98% entire home listings, and Figure 5 indicates an almost exclusive Manhattan concentration (99%), consistent with a monthly corporate or executive rental market.





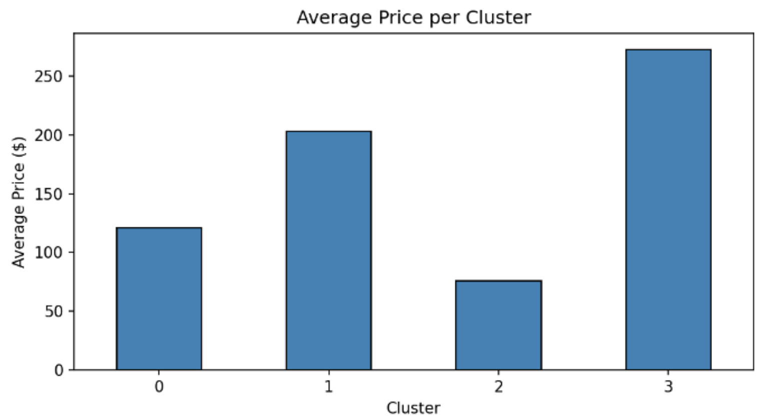


Figure 3: Average Price per Cluster A bar chart of four clusters, Cluster 0 ($121), Cluster 1 ($203), Cluster 2 ($76) and Cluster 3 ($273). Cluster 3 (Long-Stay Luxury) has the highest average price of over three times as much as Cluster 2 (Budget Private Rooms).

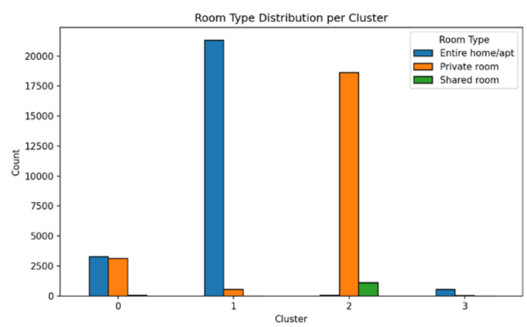


Figure 4: Distribution of Room Type by Cluster A grouped bar chart of the number of whole home/apt (blue), private room (orange), and shared room (green) facility in each cluster. Cluster 1 has a dominance of entire home (~21,000 listings), Cluster 2 has a dominance of private room (~18,500) with a visible presence of shared room (1,100) whereas Cluster 0 is almost evenly split between entire home and private room. Cluster 3 is small and nearly all of its buildings are intact homes.

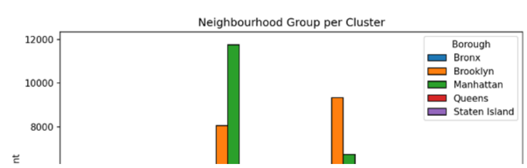


Table 1: Cluster Profile Summary

Cluster	Label	Avg Price	Dominant Room Type	Main Borough
0	Mid-Range Mixed	$121	Entire/Private (50/50)	Brooklyn 41%
1	High-Value Entire Homes	$203	Entire home (98%)	Manhattan 54%
2	Budget Private Rooms	$76	Private room (94%)	Brooklyn 47%
3	Long-Stay Luxury Apts	$273	Entire home (98%)	Manhattan 99%

4. Discussion and Recommendations

Location and room type came out as the clear pricing drivers, which matches what the literature suggests (Wang and Nicolau, 2017). Despite some prediction error was expected, Random Forest performed best of the tested models. Photos, host reputation, or seasonal demand are simply impossible to capture in structured tabular data. The clusters are interpretable and tell a coherent story about the NYC market, but the moderate silhouette scores and overlap between segments suggest the boundaries aren't as clean in reality. Airbnb could still use these profiles to guide hosts on pricing strategy, but with that, take into account the limitations. The biggest limitation is the data itself. Using more updated data with richer features would be a bigger improvements as next step (Jain, 2010).

Based on these findings, hosts in Manhattan would benefit from positioning entire home listings towards longer stays, while Brooklyn hosts could attract more budget-conscious travellers by competitively pricing private rooms.

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